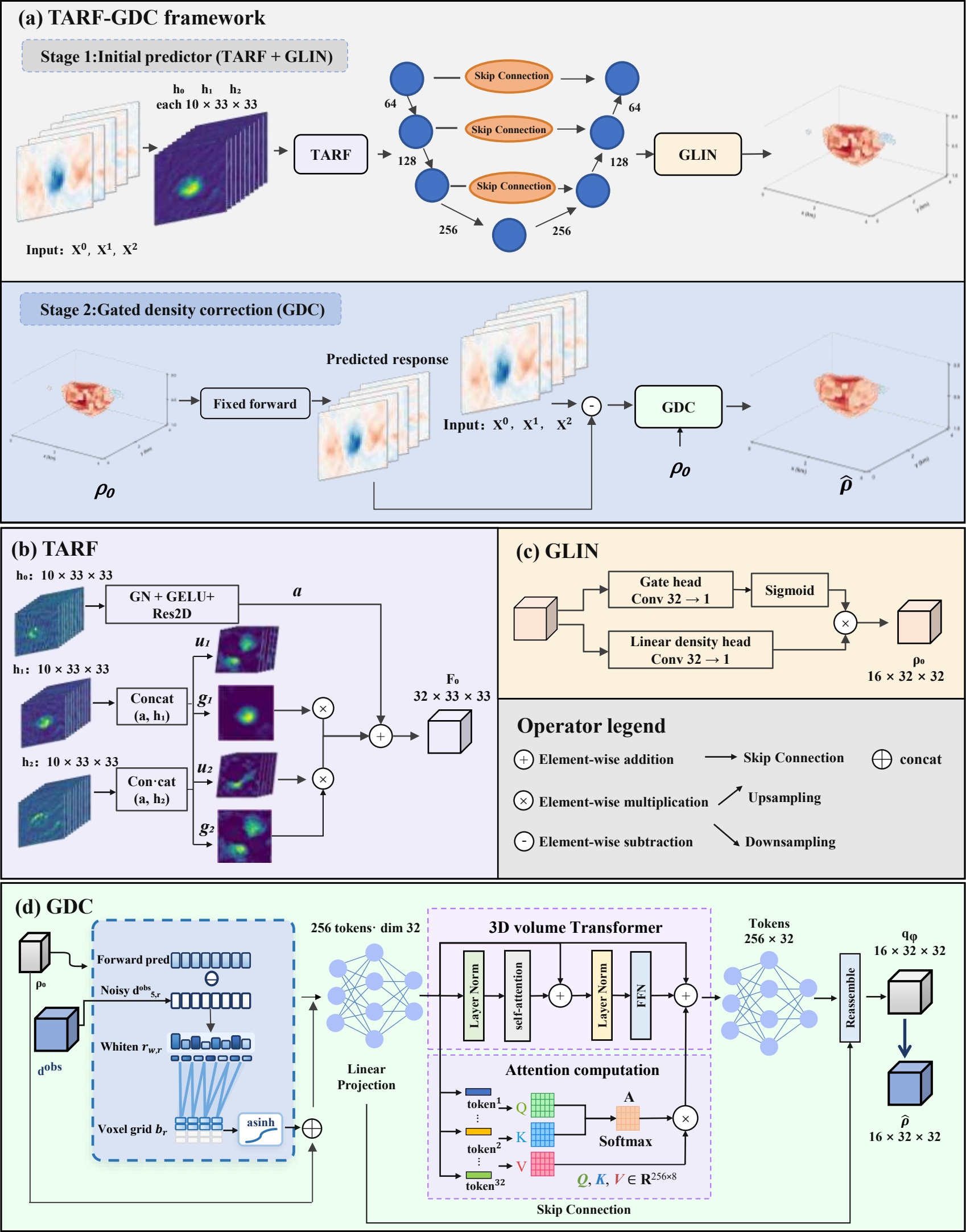

(a) TARF-GDC framework
Stage 1:Initial predictor (TARF + GLIN)
h₀ h₁ h₂
each 10 × 33 × 33
Input: X⁰, X¹, X²
TARF
64
128
256
Skip Connection
Skip Connection
Skip Connection
64
128
256
GLIN
Stage 2:Gated density correction (GDC)
Fixed forward
Predicted response
Input: X⁰, X¹, X²
GDC
ρ₀
ρ̂
(b) TARF
h₀: 10 × 33 × 33
GN + GELU+ Res2D
a
u₁
h₁: 10 × 33 × 33
Concat (a, h₁)
g₁
h₂: 10 × 33 × 33
Con·cat (a, h₂)
u₂
g₂
F₀
32 × 33 × 33
(c) GLIN
Gate head Conv 32 → 1
Sigmoid
Linear density head Conv 32 → 1
ρ₀
16 × 32 × 32
Operator legend
Element-wise addition
Skip Connection
concat
Element-wise multiplication
Upsampling
Element-wise subtraction
Downsampling
(d) GDC
ρ₀
d^obs
Forward pred
Noisy d^obs_{5,r}
Whiten r_{w,r}
Voxel grid b_r
asinh
256 tokens· dim 32
Linear Projection
3D volume Transformer
Layer Norm
self-attention
Layer Norm
FFN
Attention computation
token¹
token²
token³²
Q
K
V
A
Softmax
Q, K, V ∈ R^{256×8}
Tokens 256 × 32
Reassemble
qφ
16 × 32 × 32
ρ̂
16 × 32 × 32
Skip Connection